ECE 417/598 Midterm 2 2024

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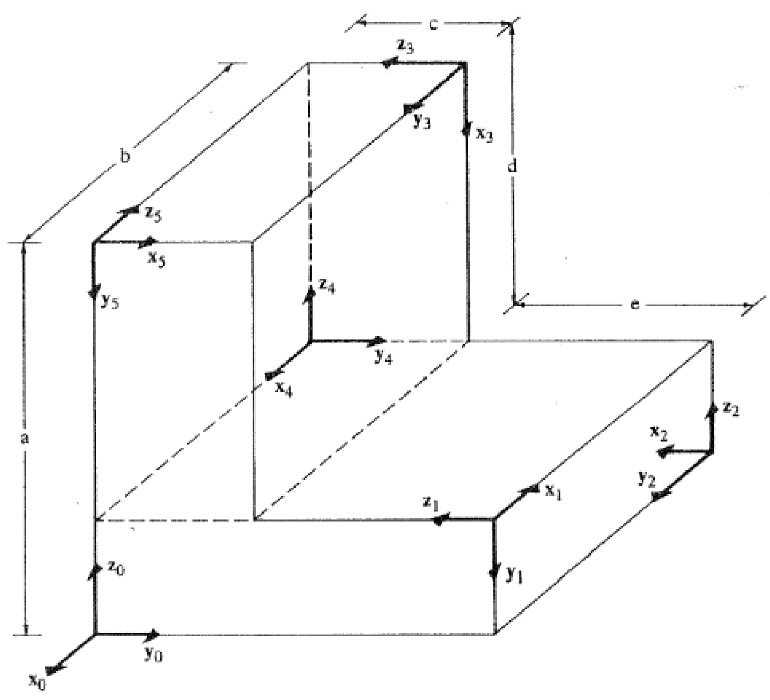
(1) Student name: Student email:

About the exam

1. There are total 5 problems. You must attempt all 5.
2. Maximum marks: 50.
3. Maximum time allotted: 50 min
4. Calculators are allowed.
5. One 8x11 in cheat sheet (both-sides) is allowed.

Problem 1 *Find the 4x4 transformation matrix* ${}^0T_2$ *that transforms coordinates from coordinate frame 2 to coordinate frame 0 (5 marks).*



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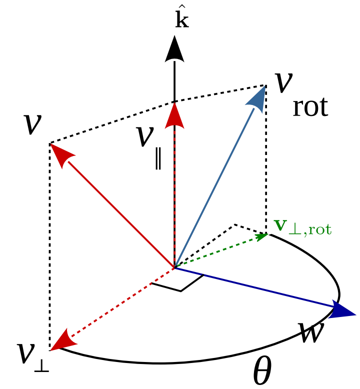
Problem 2 *In your own words, prove using trignometry that the 2D rotation matrix is given by (10 marks)*

$$R = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}. \tag{1}$$

Problem 3 *(Rodrigues formula) In the figure below, we rotate a point* $\mathbf{v}$ *around axis unit-vector* $\hat{\mathbf{k}}$ *by an angle* θ*. A unit vector* $\hat{\mathbf{w}}$ *is perpendicular to the both* $\mathbf{v}$ *and* $\hat{\mathbf{k}}$*. Another vector* $\mathbf{v}_\perp$ *is the projection of* $\mathbf{v}$ *onto a plane that is perpendicular to* $\hat{\mathbf{k}}$*. Note that* $\mathbf{v}_\perp$ *is perpendicular to both* $\hat{\mathbf{w}}$ *and* $\hat{\mathbf{k}}$ *(10 marks).*



a. *write the unit-vector* $\hat{\mathbf{w}}$ *in terms of* $\mathbf{v}$ *and* $\hat{\mathbf{k}}$ *.*

b. *Then write the vector (including the correct magnitude)* $\mathbf{v}_\perp$ *in terms of* $\mathbf{v}$ *and* $\hat{\mathbf{k}}$ *(5 marks).*

Problem 4 *Consider a unit-vector* $\hat{\mathbf{k}} = (k_x, k_y, k_z)^\top$ *and the corresponding cross-product matrix,*

$$K = [\hat{\mathbf{k}}]_\times = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}. \tag{2}$$

Using the geometry of the cross-product prove that $K^3 = -K$. *Hint on next page*. *(5 marks)*

(Hint for Problem 4: Chose any vector $\mathbf{x} \neq \hat{\mathbf{k}}$. Draw the following three vectors: (1) $\mathbf{a} = \hat{\mathbf{k}} \times \mathbf{x}$, (2) $\mathbf{b} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x})$, (3) $\mathbf{c} = \hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times (\hat{\mathbf{k}} \times \mathbf{x}))$. Compare vectors $\mathbf{c}$ and $\mathbf{a}$. What do you observe?)

Problem 5 *Derive the rotation matrix corresponding to the Euler angles* (α, β, γ) *representation* $R = R_x(\alpha)R_y(\beta)R_z(\gamma)$. *Also derive an expression to convert the rotation matrix back to Euler angles. (20 marks).*